Introduction to NLP

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Definition and Key Concepts I

- Natural Language Processing (NLP) is a branch of artificial intelligence (AI) that focuses on understanding, interpreting, and manipulating human language by machines. NLP encompasses domains such as linguistics, computer science, and AI to create tools capable of interpreting and synthesizing text for various applications.
- The goal of NLP is to enable computers to understand text and spoken words in the same way as humans do, by analyzing the grammatical structure of sentences and the individual meaning of words, and then using algorithms to extract the overall meaning of the text and provide results to user queries.

Definition and Key Concepts II

- Applications of NLP include automatic translation, voice command response, large-scale text synthesis, and natural language response generation.
- Introduction to NLP requires an understanding of key concepts underlying this field. One of the essential tasks of NLP is Automatic Speech Recognition (ASR), which involves converting speech into text. This technology is used in transcription systems, voice assistants, and many other applications like those developed by Google, Apple, or Amazon.

Definition and Key Concepts III

- Another key component of NLP is Natural Language Generation (NLG), which involves producing text or human language from structured data or previously acquired information. This can be used in applications such as automated report writing, summary generation, and interactive dialogues.
- NLP also relies on machine learning techniques such as supervised and unsupervised learning to train language models capable of understanding and generating text. These models often rely on deep neural networks, such as neural networks, which can capture the structure and relationships of language.

Definition and Key Concepts IV

- Sentiment analysis is another important application of NLP, which determines the opinion, emotion, or attitude expressed in text. This is used in the marketing field to assess customer perception of a product or brand, as well as in strategic intelligence to track public opinion on specific topics.
- Finally, NLP encompasses other areas such as automatic translation, context-based natural language understanding, Named Entity Recognition (NER) to extract specific information, to identify relationships between different parts of the text, and many others.

Definition and Key Concepts V

- NLP is a multidisciplinary field that combines techniques from AI, linguistics, and machine learning to enable machines to understand and generate human language accurately and efficiently. It plays a crucial role in various applications, from text data analysis to human-machine interaction, and continues to evolve rapidly with new technological advancements.

Definition and Key Concepts VI

In summary, NLP is important as it facilitates communication between humans and machines, allowing computers to process and analyze textual and vocal data effectively. This can help businesses automate tasks, analyze data for valuable insights, and improve the quality of analyses. Applications of NLP are numerous and include chatbots, voice assistants, search engines, automatic translation, sentiment analysis, and many more.

Applications of NLP in Various Domains I

▶ As said before, NLP offers numerous applications in various domains. Firstly, in the field of virtual assistance, NLP is used to develop intelligent personal assistants such as Siri, Alexa, and Google Assistant. These systems utilize automatic **speech recognition and language understanding** to respond to user queries, schedule events, provide information, and execute voice commands.

▶ In the field of information retrieval, NLP is used to extract relevant information from vast volumes of textual data.

Applications of NLP in Various Domains II

- NLP also finds applications in **sentiment analysis** and **opinion mining**. It enables the analysis of sentiments and opinions expressed in social media, product reviews, user comments, etc.
- NLP is also used in the finance domain, where it enables the analysis of financial reports, market trends tracking, fraud detection, and automation of financial document processing. Engineers can apply NLP to develop financial analysis and language-based forecasting systems.

Applications of NLP in Various Domains III

Text Classification	Machine Translation
Text Similarity	Text Summarization
Information Extraction	Information Retrieval
Chat Bots	Voice Assistants

Table – NLP Use Cases

Applications of NLP in Various Domains IV

In summary, NLP offers exciting applications in domains such as virtual assistance, information retrieval, sentiment analysis, healthcare, finance, and education. Researchers and engineers can leverage NLP techniques to develop innovative systems and contribute to advancements in various sectors through artificial intelligence and human language analysis.

Foundations of NLP I

NLP is a multidisciplinary field where several specialized domains interact to build an operational and efficient NLP model.

The foundations of NLP are based on computational linguistics, statistics applied to NLP, and computer science. A deep understanding of these concepts is essential for developing effective and innovative NLP systems.

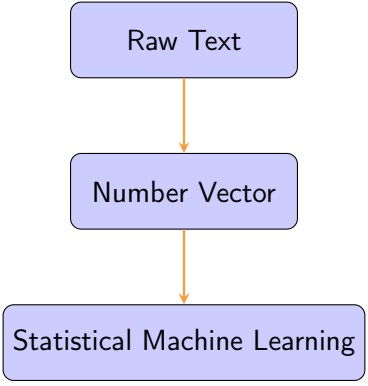
NLP Techniques I

There are 3 large NLP Techniques :

Rules & Heuristics	Machine Learning	Deep Learning

Table – Three category NLP Techniques

Rules and Heuristics category



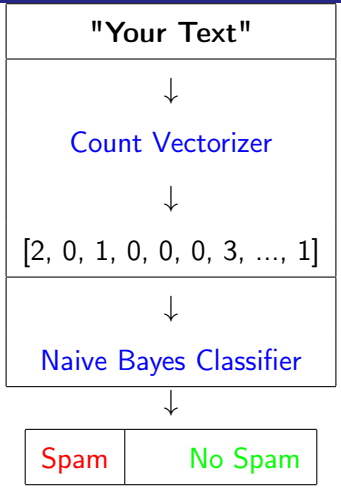
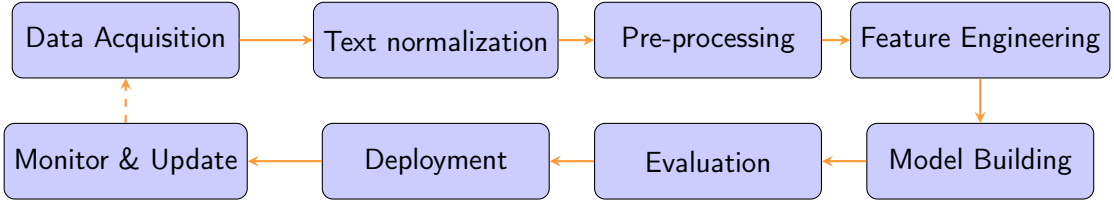


Table – Example

NLP Pipeline I

The general and complete NLP pipeline is the following,



Text Normalization I

Text normalization in NLP consists in transforming the text into a unique canonical form.

Some commonly used text normalization techniques in NLP :

- Converting to Lowercase
- Removing Accents
- Expanding Contractions (e.g., "I've" to "I have")
- Spelling Correction

Text Normalization II

- Removing Special Characters such as emoticons and symbols

Text normalization is an important step in NLP data preprocessing as it reduces the variability of words and facilitates subsequent analyses and modeling. Once the text is normalized, it can undergo further preprocessing steps such as tokenization, stop word removal, stemming, and lemmatization.

Pre-processing. Text Vectorization I

Text vectorization is a fundamental step in Natural Language Processing (NLP) that allows representing textual data in the form of **numerical vectors**, understandable by machine learning algorithms. This transformation is necessary because machine learning models cannot directly process textual data.

There are several approaches to text vectorization in NLP. One of the most commonly used methods is the **bag-of-words** representation.

In this approach, each document is represented as a vector whose dimensions correspond to the words present in the corpus.

Pre-processing. Text Vectorization II

Each element of the vector represents the frequency of occurrence of the corresponding word in the document. However, this method ignores the order of words and does not capture the relationships between them.

A more advanced approach is the **weighted bag-of-words representation (TF-IDF)**. This method takes into account the frequency of occurrence of words in a specific document, as well as their inverse frequency in the entire corpus.

This allows assigning greater weights to words specific to a document and reducing the importance of frequent and non-discriminating words.

Pre-processing. Text Vectorization III

Another popular method is the word embeddings representation, such as **Word2Vec** and **GloVe**. These techniques use machine learning models to create dense vector representations of words.

Each word is represented by a vector of real numbers, where similar words are close in the vector space.

This approach captures the semantic and syntactic relationships between words and allows capturing richer information about the text.

Pre-processing. Text Vectorization IV

More recently, pre-trained language models, such as **BERT** (Bidirectional Encoder Representations from Transformers), have revolutionized text vectorization. These models are trained on vast textual corpora and are capable of generating vector representations of words, sentences, and even entire documents. The vector representations obtained from these models contain high-level semantic and syntactic information, which allows achieving remarkable performance in many NLP tasks.

Pre-processing. Text Vectorization V

Text vectorization in NLP plays an essential role in various tasks, such as text classification, sentiment analysis, information retrieval, and text generation. It allows transforming text into a numerical format suitable for machine learning, thus facilitating the analysis and extraction of knowledge from textual data. By choosing the appropriate vectorization method, taking into account the specifics of the task and the text corpus, it is possible to obtain effective vector representations that capture the relevant information for text analysis.

High-level overview of the spaCy library I

- spaCy is an open-source Python library for modern NLP.
- spaCy comes with pre-trained language models and word vectors for over 60 languages.
- spaCy is focused on production and delivering code, unlike its more academic predecessors.
- The most famous and frequently used Python predecessor is NLTK. The main goal of NLTK was to give students and researchers an idea of language processing. It never claimed to be efficient.

High-level overview of the spaCy library II

- spaCy focused on providing production-ready code from day one. You can expect the models to work on real-world data, the code to be efficient and capable of processing a huge amount of text data in a reasonable time.

You can find spaCy's features and performances on its website. Here is a simple benchmark

High-level overview of the spaCy library III

Name	Spark NLP	spaCy	NLTK	CoreNLP
Sentence detection	Yes	Yes	Yes	Yes
Tokenization	Yes	Yes	Yes	Yes
Stemming	Yes	Yes	Yes	Yes
Lemmatization	Yes	Yes	Yes	Yes
POS tagger	Yes	Yes	Yes	Yes
NER	Yes	Yes	Yes	Yes
Dependency parse	Yes	Yes	Yes	Yes
Text matcher	Yes	**Yes**	**No**	Yes
Date matcher	Yes	**No**	**No**	Yes
Chunking	Yes	Yes	Yes	Yes
Spell checker	Yes	**No**	**No**	**No**
Sentiment detector	Yes	**No**	**No**	Yes
Pretrained models	Yes	Yes	Yes	Yes

▶ spaCy's code is also professionally maintained.

▶ You can also check out spaCy's GitHub repository at https ://github.com/explosion/spaCy, report a bug, or ask for help from the community.

▶ Another predecessor is CoreNLP (also known as StanfordNLP). CoreNLP is implemented in Java. Although CoreNLP is competitive in terms of efficiency, Python won thanks to the ease of prototyping and spaCy is much more professional as a software package.

▶ We will use the latest versions of spaCy for all our computational linguistics and machine learning needs.

Here are the features of the latest version :

- Tokenization preserving original data.
- Statistical sentence segmentation.

- Named Entity Recognition (NER)
- Part-of-Speech (POS) tagging.
- Dependency parsing.
- Pre-trained word vectors.
- Easy integration with popular deep learning libraries. PyTorch, TensorFlow, and MXNet.

Installation of spaCy I

- Installation of spaCy with pip

  You can install spaCy with the following command :

```
pip install spacy
```

- Updating spaCy version to the latest

```
pip install -U spacy
```

- Installation of spaCy with conda The conda support is provided by the conda community. The command to install spaCy with conda is the following :

```
conda install -c conda-forge spacy
```

- Checking the version you have :

```
python -m spacy info
```

Installation of spaCy on Windows

▶ If you have a Windows system, you need to install a version of Visual C++ Build Tools or Visual Studio Express that matches your Python distribution.

You should be able to import spaCy into your Python shell with the command : `import spacy`

Basic Operations with spaCy I

- In this chapter, you will learn the basic operations with spaCy, such as creating a language pipeline, tokenizing the text, and splitting the text into sentences.

First, you will learn what a language processing pipeline is and the components of the pipeline. We will continue with the general conventions of spaCy - important classes and organization of classes - to help you better understand the organization of the spaCy library and develop a solid understanding of the library itself.

You will then learn the first component of the pipeline : the Tokenizer.

Basic Operations with spaCy II

You will also learn an important linguistic concept : lemmatization, as well as its applications in Natural Language Understanding (NLU).

Then, we will cover in detail the container classes and data structures of spaCy.

We will end the chapter with useful features of spaCy that you will use in daily NLP development.

Overview of spaCy conventions I

Every NLP application consists of several steps of text processing. As you can see in the first chapter, we have always created instances called `nlp` and `doc`. But what exactly did we do ?

When we call `nlp` on our text, spaCy applies certain processing steps. The first step is tokenization to produce a `Doc` object. The `Doc` object is then further processed with a tagger, a parser, and an entity recognizer. This way of processing text is called a language processing pipeline. Each pipeline component returns the processed `Doc` and then passes it to the next component :

Overview of spaCy conventions II

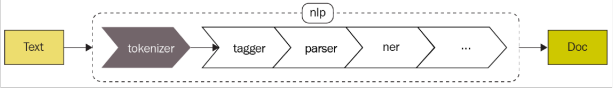


Figure – High-level view of the processing pipeline